To get some sense of just how far Tau Ceti *is* compared to more familiar things, let's scale its distance down enormously. Imagine it as the distance from New York City to Perth, Australia, about halfway around the world.

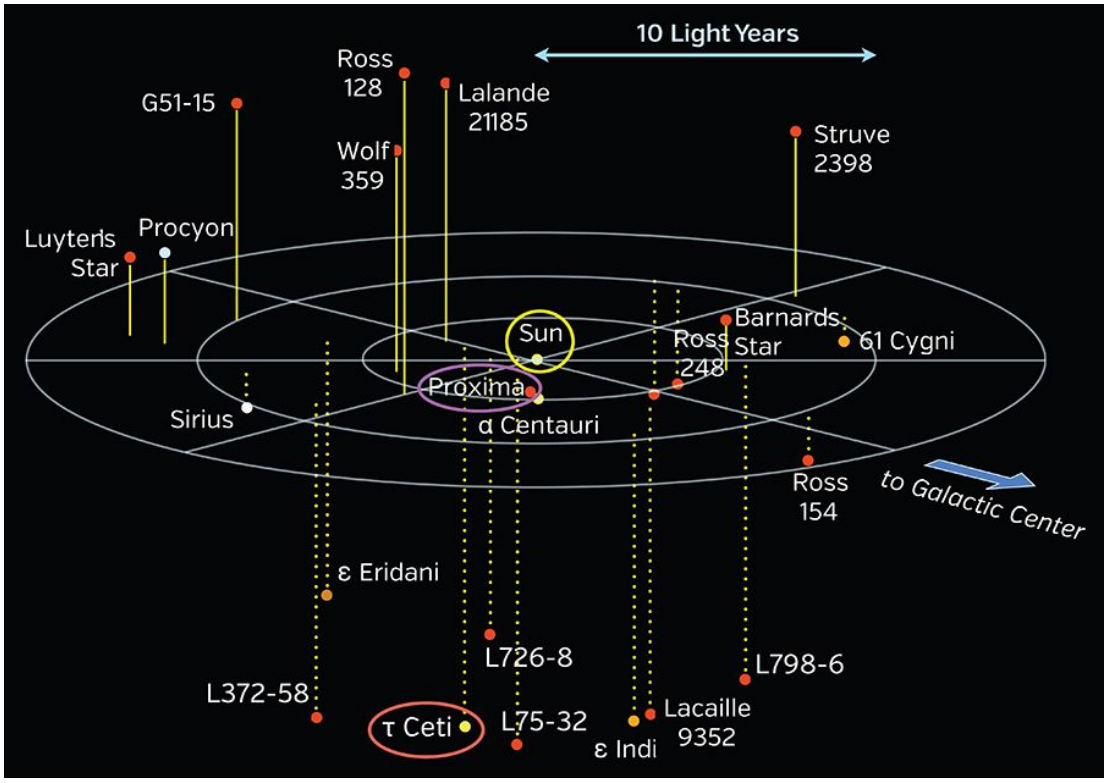


Fig. 13.1. All the stars within 12 light-years of Earth. The Sun, Proxima Centauri, and Tau Ceti are circled in yellow, purple, and red. *[I adapted this map from Richard Powell's www.atlasoftheuniverse.com.]*

The very nearest star other than the Sun is Proxima Centauri, 4.24 light-years from Earth, but there is no evidence it has habitable planets. With Tau Ceti's distance imagined as New York to Perth, then Proxima Centauri's is like New York to Berlin. It's not a lot closer than Tau Ceti!

For comparison, the most distant unmanned spacecraft that humans have sent into interstellar space is *Voyager 1*, now about 18 light-hours from Earth. It has been traveling for thirty-seven years to get there. With Tau Ceti's distance imagined as New York to Perth, then Earth to